Equilibrium and missing forces

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find a missing force when acceleration is zero.

SUCCESS CRITERIA - I can...

1. Recognize rest and constant velocity as possible equilibrium.
2. Find missing forces when the net force is zero.
3. Explain why turning is not equilibrium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “equilibrium” and “balanced” to explain your answer. Sentence starters are optional.

equilibrium: zero net force and zero acceleration

balanced: adding to zero net force in this force context

tension: the pulling force through a stretched rope or cable

01 Warm-up: bring back an earlier idea

A **3 kg** box slides left at **5 m/s** on a rough horizontal floor after release. **Draw forces** and **explain what follows about vertical** and horizontal acceleration.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

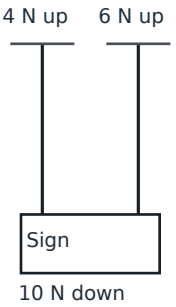
02 Worked example: explain the prediction

Reference: $\Sigma F_x = \mathbf{0}$; $\Sigma F_y = \mathbf{0} \Rightarrow a = \mathbf{0}$.
Equilibrium allows rest or constant velocity. Resolve missing forces along each axis. Use g = **9.8 N/kg**.

Worked reasoning: A sign has weight **10 N** down and two vertical upward pulls of **4** and **6 N**. The vector sum is zero, so equilibrium is possible.

Your turn: 1. Draw and **label all three forces**.
2. **Name the equation** that checks vertical balance.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

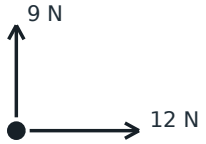
Sentence starters are optional. You may plan in any language.

03 Two component balance

A stationary object has **12 N** right and **9 N** up. A third force balances both. **Find its components**, magnitude and direction.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



The dot represents the object. Third force not shown

Marzano check - circle: 1 2 3

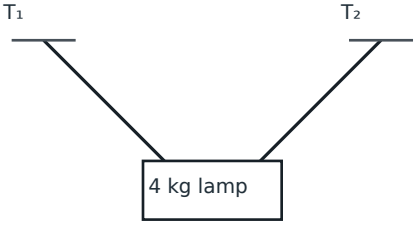
04 Two-rope hanger

A stationary **4 kg** lamp hangs from two identical ropes, each **45°** above horizontal. Use g = **9.8 N/kg**. Resolve the tensions into components and **find each tension**. **Explain why each tension** is larger than half the weight.

Word help: resolve = separate a vector into components

Sentence starter: Both … . They differ in … because … .

45° above horizontal



Marzano check - circle: 1 2 3

05 Change angle

A **10 kg** sign hangs from two identical ropes.
1. **Compare ropes 30°** above horizontal with ropes **60°** above horizontal.
2. Use g = **9.8 N/kg**.
3. **Sketch both arrangements**, calculate each tension, and **explain which puts more tension** in the ropes.

Sentence starter: Both … . They differ in … because … .

First arrangement: 30°



Marzano check - circle: 1 2 3

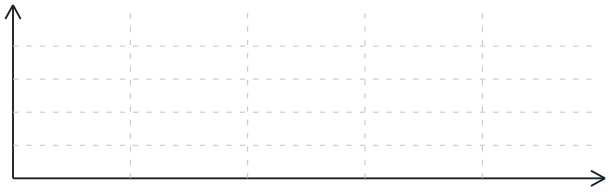
06 Review: connect an earlier idea

A=(**7**, **-5**) m and B=(**-5**, **9**) m. **Find R**, magnitude and direction. **Choose C** so A+B+C=**0** and **sketch the route**. **Draw the component vectors** and resultant; **label lengths**.

Word help: component = the part of a vector along one chosen axis

Sentence starter: I found … [unit]. This value means … in this situation.

North / +y (m)



East / +x (m)

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A **3 kg** lamp hangs at rest from a vertical rope. Use g = **9.8 N/kg**. **Find tension**. Then redraw the forces if the lamp accelerates upward. **Write the equation showing why** tension must change.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Add and label force arrows.

Choose the lamp as the object.

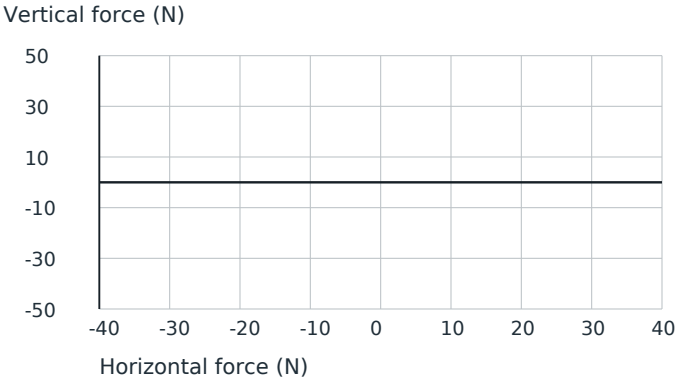
Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A stationary junction has a **30 N** horizontal force right, a **40 N** vertical force down, and one unknown force. **Find the unknown vector**, magnitude and direction.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3